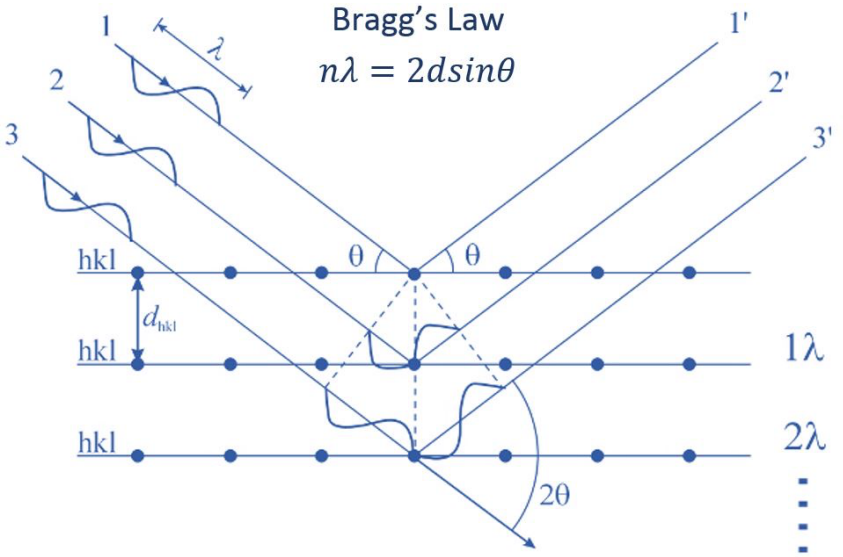
# Fundamental Physics II
# Optics - Chapter 1

For B1 Students
Academic year 2024-2025

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# OVERVIEW

| Chapter | Contents | Hours | | | Ref./Resources | Assignment(s) |
|---|---|---|---|---|---|---|
| | | Lect. | Exr. | Prc. | | |
| 1 | The nature and propagation of light | 2.0 | 1.0 | | | Exercises |
| 2 | Geometric optics | 5.0 | 2.0 | | | Exercises |
| 3 | Interference | 3.0 | 1.5 | | | Exercises |
| 4 | Diffraction | 3.0 | 1.5 | | | Exercises |
| | Midterm test (0.5h) | | | | | |
| 5 | Photons: Light waves behaving as particles* | | | | | |
| | | 13 | 6 | | | |

Textbook:

[1] Young and Freedman – University Physics with Modern Physics 15th Edition (2020)

**Suggested exercises** (in the textbook):

**33.** 1, 2, 7, 11, 17, 23, 25, 27, 31, 35, 37, 41

**34.** 3, 5, 8, 13, 19, 21, 26, 29, 31, 32, 40, 43, 45, 50, 52, 54, 58, 59, 62, 63, 66, 71

**35.** 2, 3, 4, 6, 9, 11, 14, 17, 19, 26, 35

**36.** 1, 6, 8, 14, 15, 19, 21, 26, 31, 33, 36, 37, 39, 42, 43, 44, 45

| Assessment/ Evaluation | | |
|---|---|---|
| | **Attendance/Attitude** | **10%** |
| | Exercise(s) | 0% |
| | Practicals | 0% |
| | **Mid-term test** | **30 %** |
| | **Final exam** | **60 %** |

# Chapter 1. The nature and propagation of light (2.0h)

1.1 The nature of light

1.2 Reflection and refraction

1.3 Total internal reflection

1.4 Dispersion

1.5 Polarization

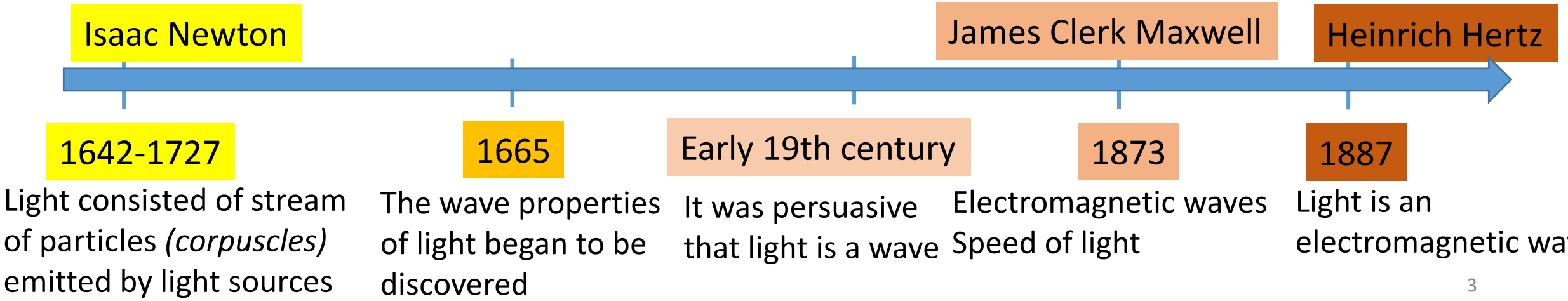
1.6 Scattering of light

1.7 Huygen's principle

**List of Exercises:**

In textbook "University Physics with Modern Physics" 15th Edition, Hugh D. Young, Roger A. Freedman, A. Lewis Ford (2020)

**33. 1, 2, 7, 11, 17, 23, 25, 27, 31, 35, 37, 41**



# Chapter 1. The nature and propagation of light (2.0h)

## 1.1 The nature of light

### 1.1.1 The two personalities of light

**Wave property: Propagation of light**

**Particle property: Emission, Absorption**

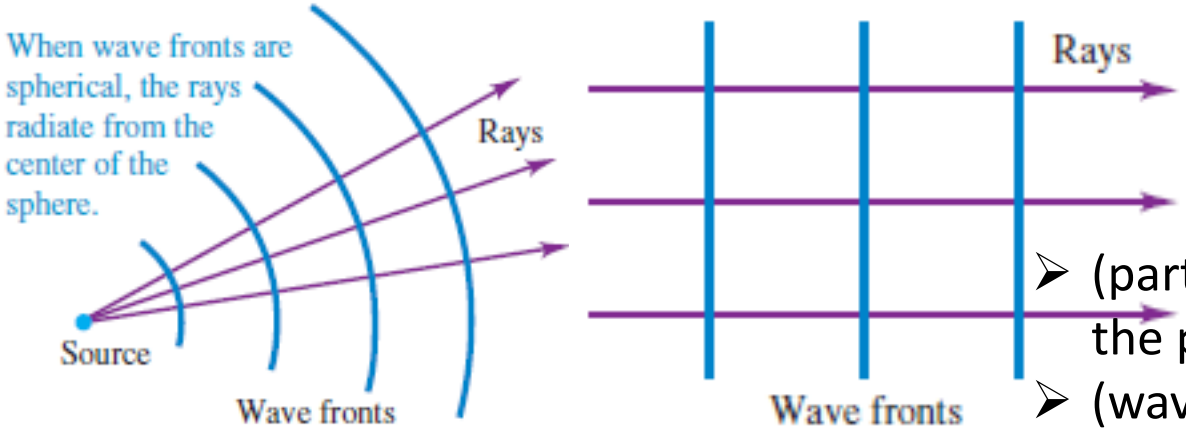
The speed of light in vacuum

$$c = 2.99792458\times10^{8}\ \text{m/s} \approx 3.00\times10^{8}\ \text{m/s}$$

An electric heating element emits primarily IR radiation. When its temperature is high enough, it also emits a discernible amount of visible light.

### 1.1.2 Waves, Wave Fronts, and Rays

**Wave Front** The locus of all adjacent points at which the phase of vibration of a physical quantity associated with the wave is the same.

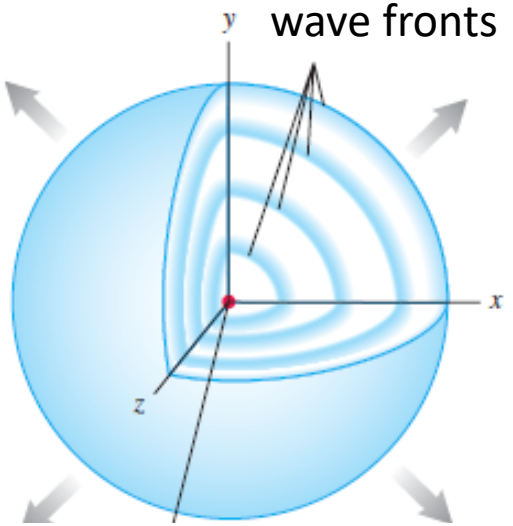



When wave fronts are planar, the rays are perpendicular to the wave fronts and parallel to each other.

**Rays**

- (particle theory of light): the paths of the particles
- (wave viewpoint): an imaginary line along the direction of travel of the wave

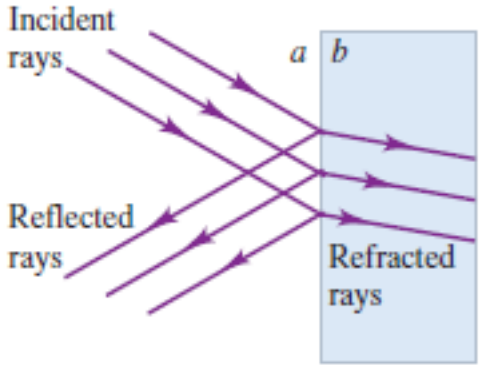
Expanding wave fronts

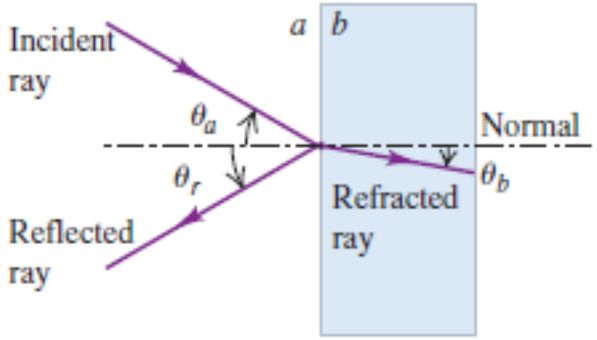


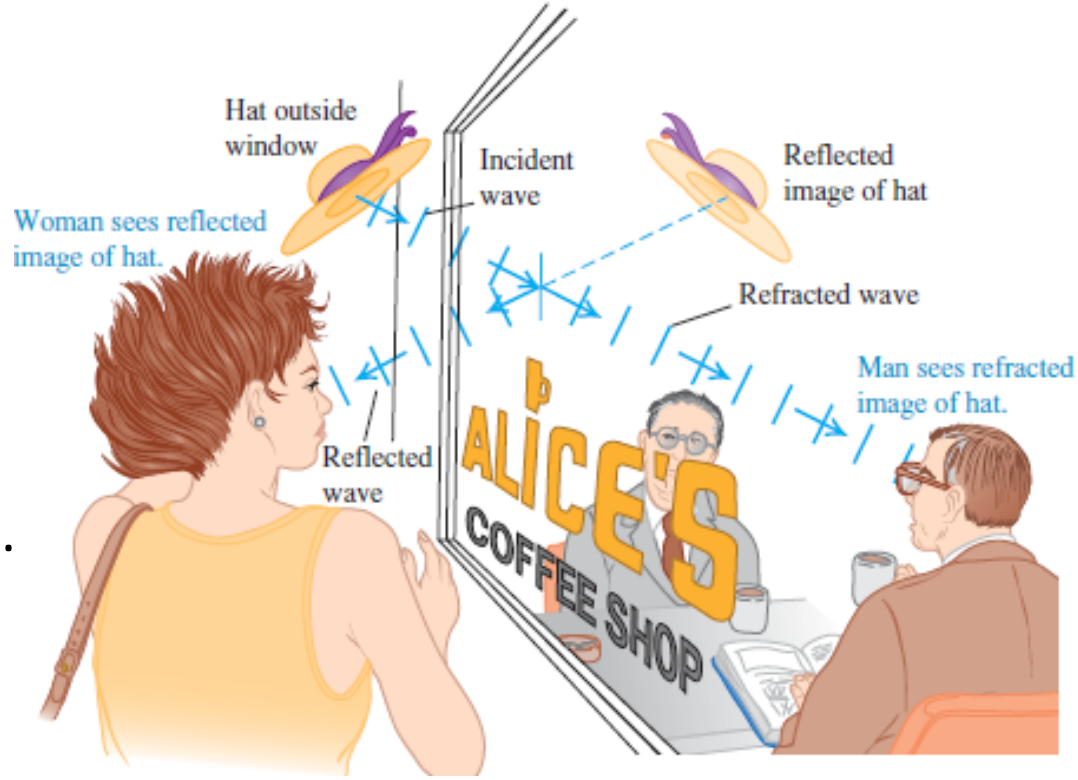

Point sound source producing spherical sound waves (alternating compression and rarefaction of air)

# Chapter 1. The nature and propagation of light (2.0h)

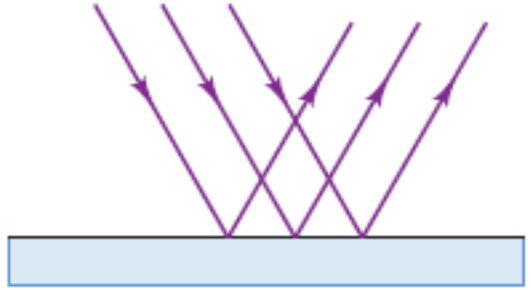
## 1.2 Reflection and refraction



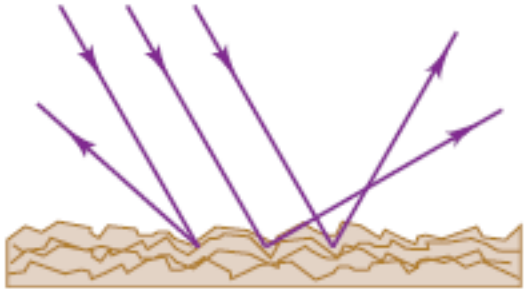




The waves in the outside air and glass represented by rays.



Specular reflection:
smooth surface



Diffuse reflection:
rough surface

$$n = \frac{c}{v}$$

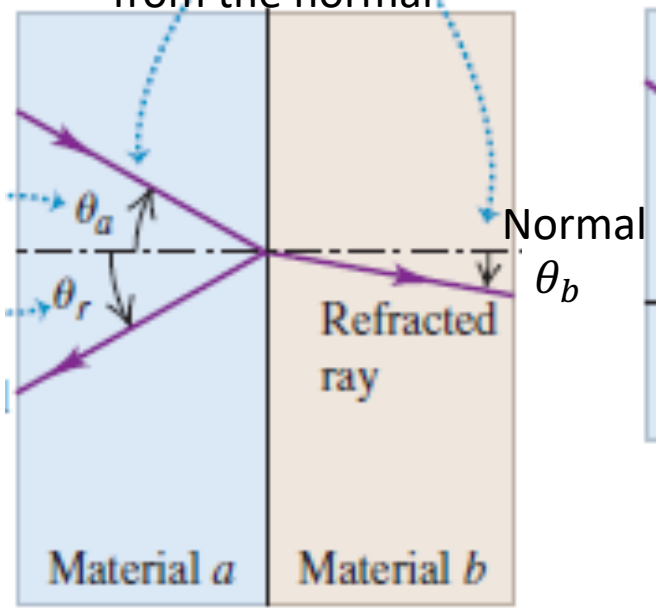
**Index of refraction** of an optical material (**refractive index**):
The ratio of the speed of light *c* in vacuum to the speed *v* in the material.

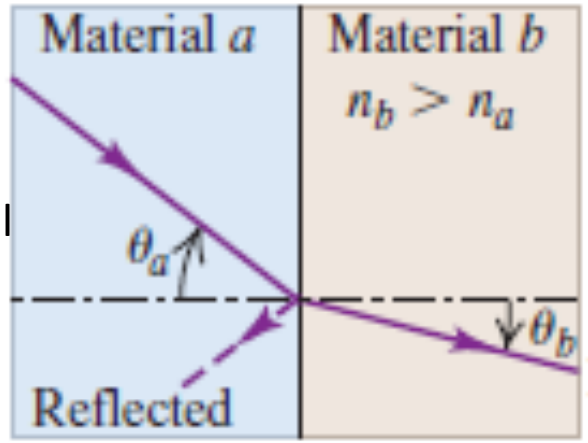
# Chapter 1. The nature and propagation of light (2.0h)

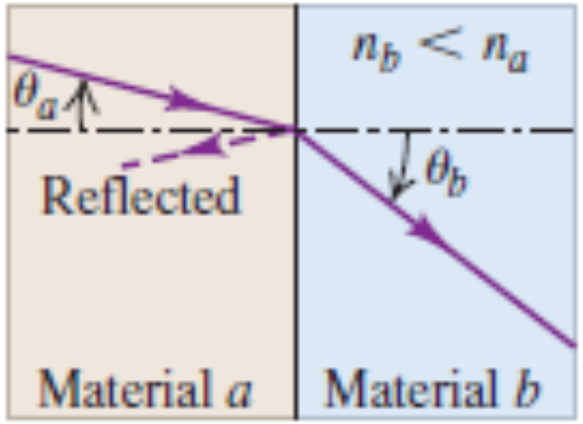
## 1.2 Reflection and refraction

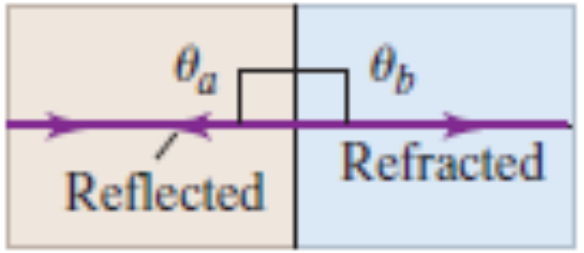
### 1.2.1 The laws of reflection and refraction

$\theta_a$ and $\theta_b$ are measured from the normal.




$$\frac{\sin\theta_a}{\sin\theta_b} = \frac{n_b}{n_a}$$

$$n_a \sin\theta_a = n_b \sin\theta_b$$ (law of refraction – **Snell's** law)

- ✓ The incident, reflected, and refracted rays and the normal to the surface all lie in the same plane
- ✓ The angle of reflection $\theta_r$ is equal to the angle of incidence $\theta_a$ for all wavelengths and for any pair of materials.
- ✓ The ratio of the sines of the angles $\theta_a$ and $\theta_b$, where both angles are measured from the normal to the surface, is equal to the inverse ratio of the two indexes of refraction.

$\theta_r = \theta_a$ (law of reflection)

# Chapter 1. The nature and propagation of light (2.0h)

## 1.2 Reflection and refraction

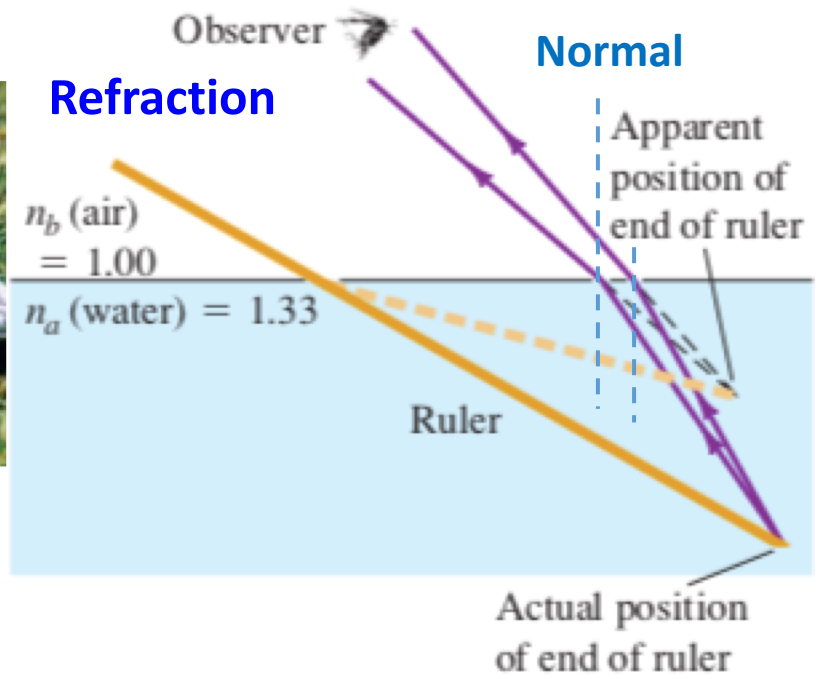
### 1.2.1 The laws of reflection and refraction

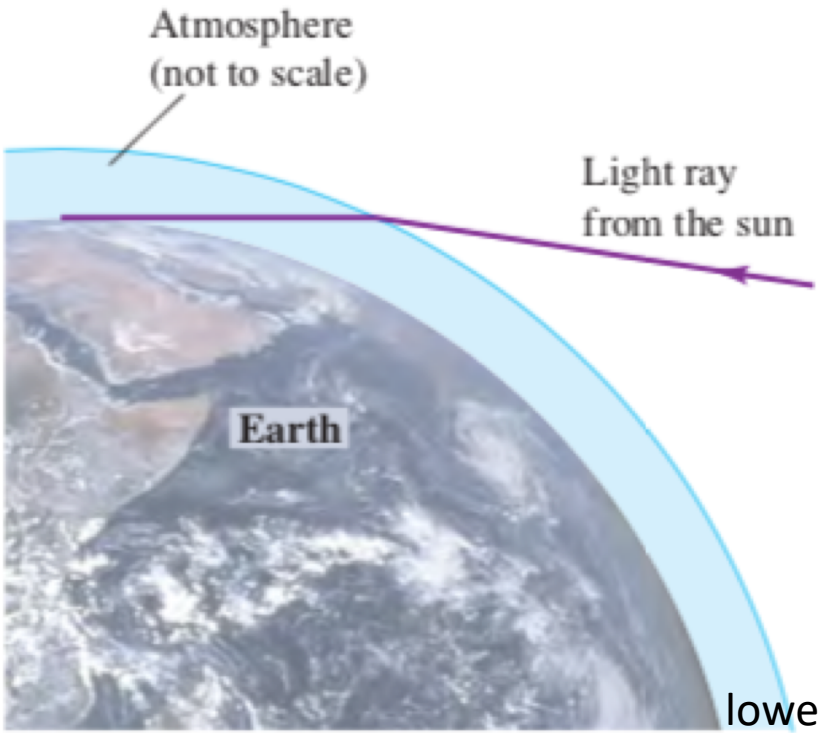
**Table 33.1 Index of Refraction for Yellow Sodium Light, $\lambda_0 = 589$ nm**

| Substance | Index of Refraction, $n$ |
|---|---|
| Solids | |
| Ice ($H_2O$) | 1.309 |
| Fluorite ($CaF_2$) | 1.434 |
| Polystyrene | 1.49 |
| Rock salt (NaCl) | 1.544 |
| Quartz ($SiO_2$) | 1.544 |
| Zircon ($ZrO_2 \cdot SiO_2$) | 1.923 |
| Diamond (C) | 2.417 |
| Fabulite ($SrTiO_3$) | 2.409 |
| Rutile ($TiO_2$) | 2.62 |
| Glasses (typical values) | |
| Crown | 1.52 |
| Light flint | 1.58 |
| Medium flint | 1.62 |
| Dense flint | 1.66 |
| Lanthanum flint | 1.80 |
| Liquids at 20°C | |
| Methanol ($CH_3OH$) | 1.329 |
| Water ($H_2O$) | 1.333 |
| Ethanol ($C_2H_5OH$) | 1.36 |
| Carbon tetrachloride ($CCl_4$) | 1.460 |
| Turpentine | 1.472 |
| Glycerine | 1.473 |
| Benzene | 1.501 |
| Carbon disulfide ($CS_2$) | 1.628 |

A ruler immersed partly in water

The ruler seems bent?



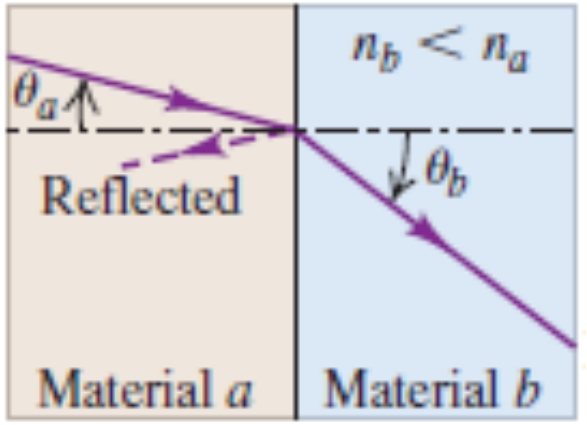

lower ➔ denser

The setting sun appears flattened vertically!

# Chapter 1. The nature and propagation of light (2.0h)

## 1.2 Reflection and refraction

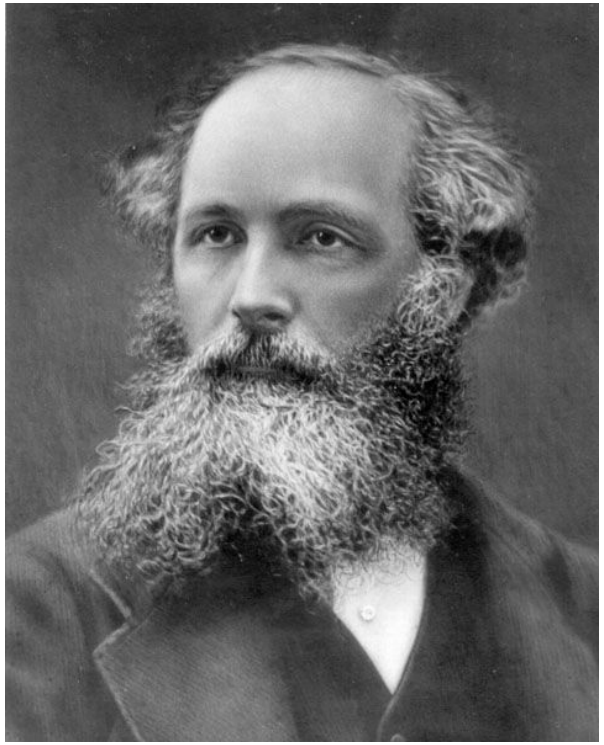
### 1.2.1 The laws of reflection and refraction




- The path of a refracted ray, reflected wave is reversible.
  - ✓ If $n_a > n_b$ => the ray is bent **away from** the normal,
  - ✓ If $n_a < n_b$ => the ray is bent **toward** the normal,

- The intensities of the reflected and refracted rays depend on:
  - ✓ The angle of incidence,
  - ✓ The two indexes of refraction,
  - ✓ The polarization.

Maxwell's equations



James Clerk Maxwell
(1831-1879)

### 1.2.2 Index of refraction and the wave aspects of light

$f = \text{const}$ the frequency f of the wave does not change when passing from one material to another

Wavelength of light in a material:

$$\lambda = \frac{\lambda_0}{n}$$

- ✓ $\lambda$ of light in a material
- ✓ $\lambda_0$ of the same light in vacuum
- ✓ $n = \frac{c}{v}$ : index of refraction (refractive index)

# Chapter 1. The nature and propagation of light (2.0h)

## 1.2 Reflection and refraction

**Exp.1.1: Reflection and Refraction**
**In this following figure, material *a* is water and material *b* is glass with index of refraction 1.52.**
**The incident ray makes an angle of 60.0° with the normal.**
**Find the directions of the reflected and refracted rays.**
**Find the angle between the reflected and refracted rays?**

- The angle of reflection is equal to the angle of incidence:

$$\theta_r = \theta_a = 60.0^\circ$$

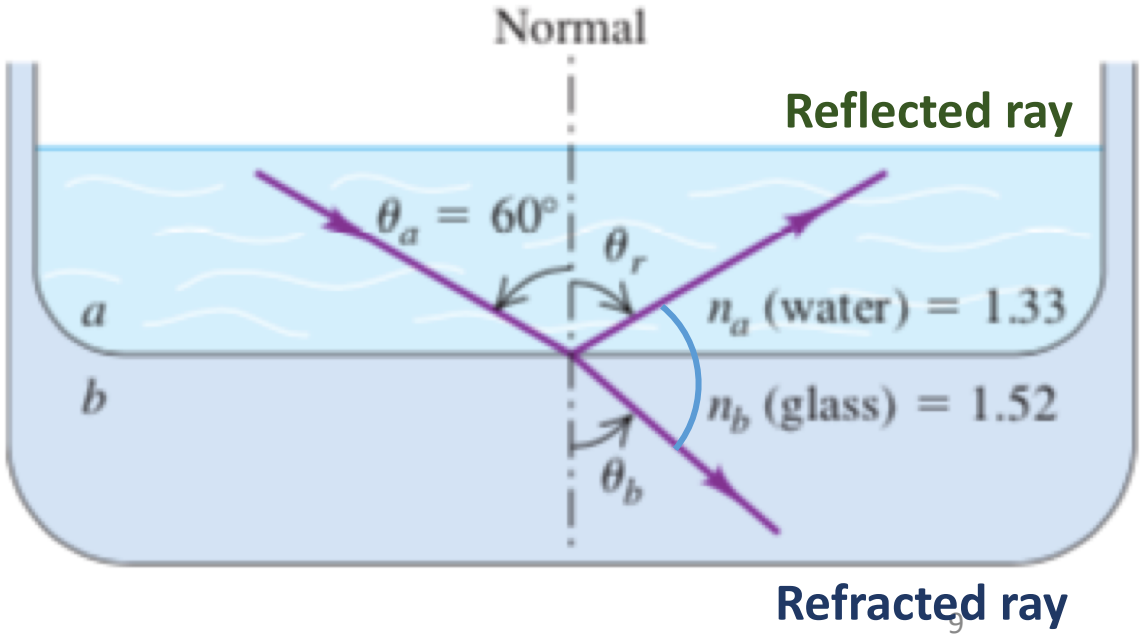
- Applying Snell's law, then the angle of refraction:

$$\sin\theta_b = \frac{n_a}{n_b}\sin\theta_a = \frac{1.33}{1.52}\sin 60.0^\circ = 0.758$$

$$\Rightarrow \theta_b = 49.3^\circ$$

- The angle between the reflected and refracted rays:

$$180^\circ - \theta_r - \theta_b = 70.7^\circ$$



# Chapter 1. The nature and propagation of light (2.0h)

## 1.2 Reflection and refraction

**Exp.1.2: Index of refraction in the eye**
**The wavelength of the red light from a helium-neon laser is 633 nm in air but 474 nm in the aqueous humor inside your eyeball.**
**Calculate the index of refraction of the aqueous humor and the speed and frequency of light in it.**

- Assume that the wavelength $\lambda_0$ in vacuum is the same as that in air.

The refractive index of the aqueous humor is:

$$n = \frac{\lambda_0}{\lambda} = \frac{633}{474} = 1.34$$

- The speed of light in the aqueous humor:

$$v = \frac{c}{n} = \frac{3.00\times10^8}{1.34} = 2.24\times10^8(\mathrm{m/s})$$

- The frequency of light in the aqueous humor :

$$f = \frac{v}{\lambda} = \frac{2.24\times10^8}{474\times10^{-9}} = 4.73\times10^{14}(\mathrm{Hz})$$

# Chapter 1. The nature and propagation of light (2.0h)

## 1.3 Total internal reflection (TIR)

- **Phenomenon:**

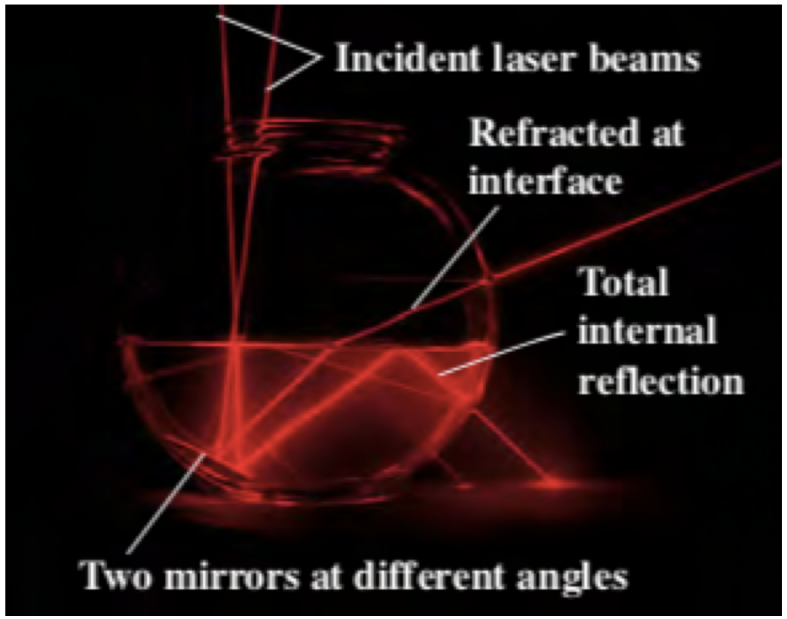
All of the light can be reflected back from the interface.

- **Condition:**

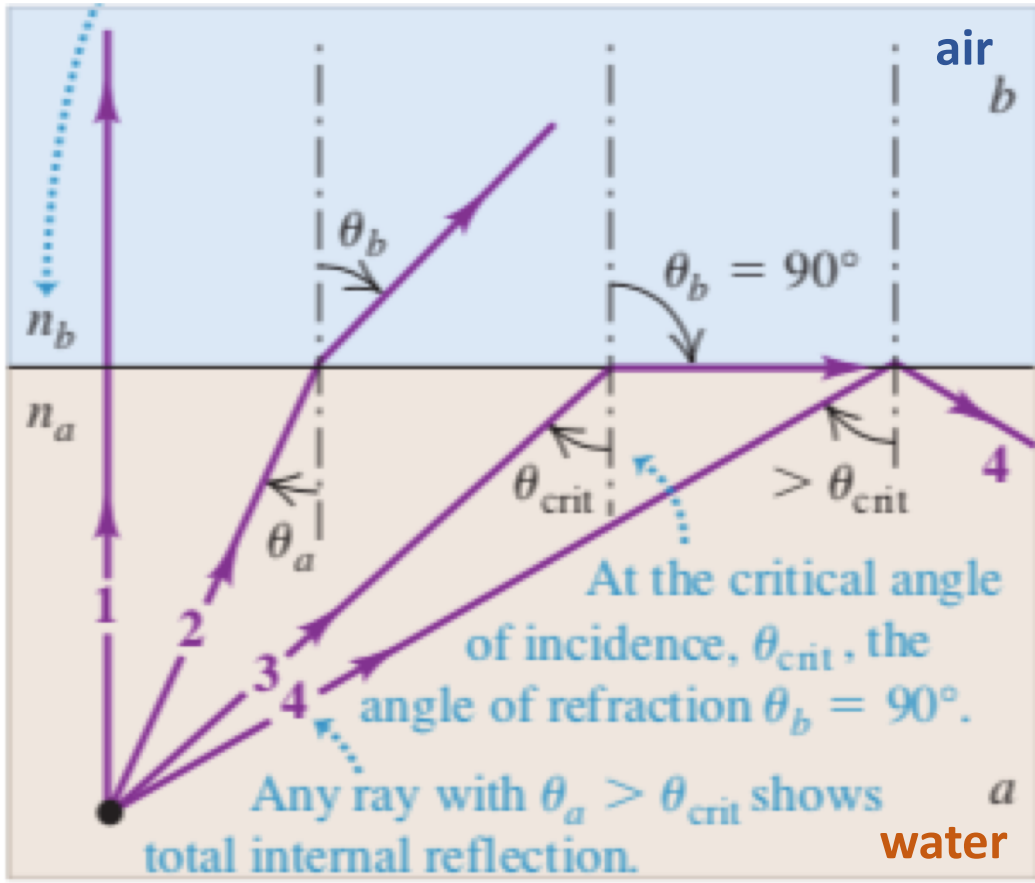
➢ $n_a > n_b$

➢ Incident angle ($\theta_a$) is larger than or equal to critical angle for total internal reflection ($\theta_{\text{crit}}$)

$$\sin\theta_{\text{crit}} = \frac{n_b}{n_a}$$




TIR occurs only if $n_b < n_a$

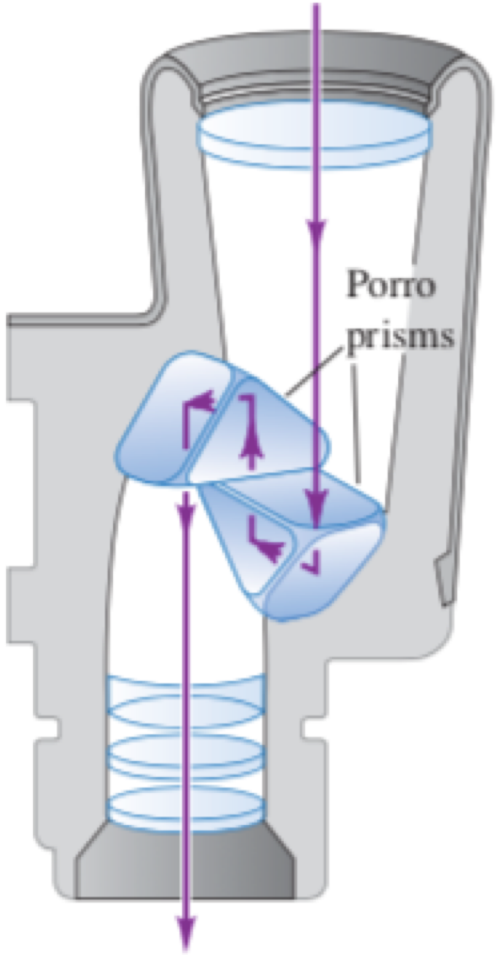


# Chapter 1. The nature and propagation of light (2.0h)

## 1.3 Total internal reflection (TIR)

- **Applications:**

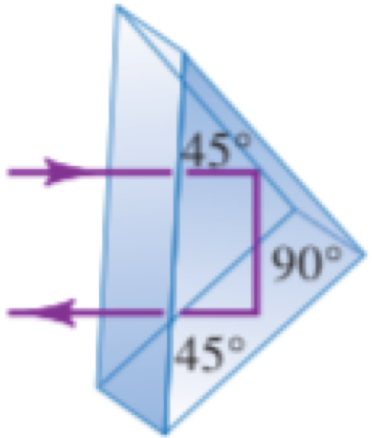
✓ Porro prism ✓ Endoscope ✓ Fiber optics




Glass-air interface:

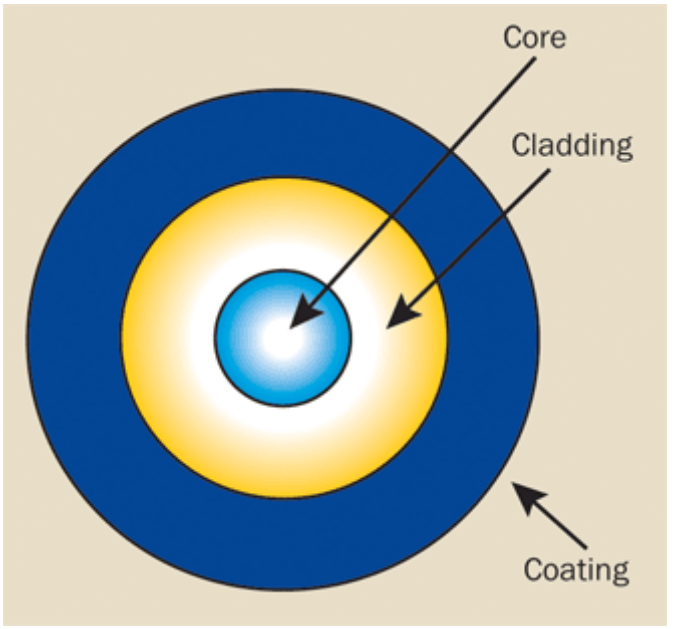
$$\sin\theta_{crit} = \frac{1}{1.52} = 0.658$$

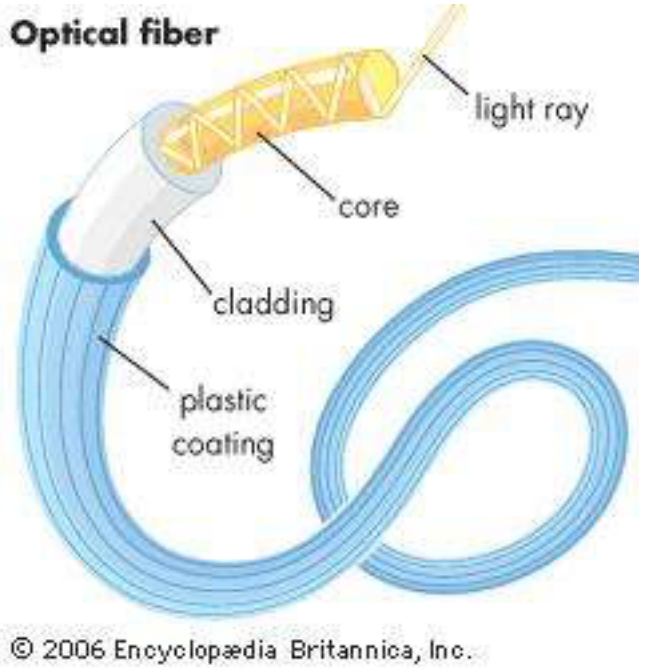
$$\theta_{crit} = 41.1^\circ$$




45° - 45° - 90° prism

Binoculars use Porro prisms to reflect the light to each eyepiece



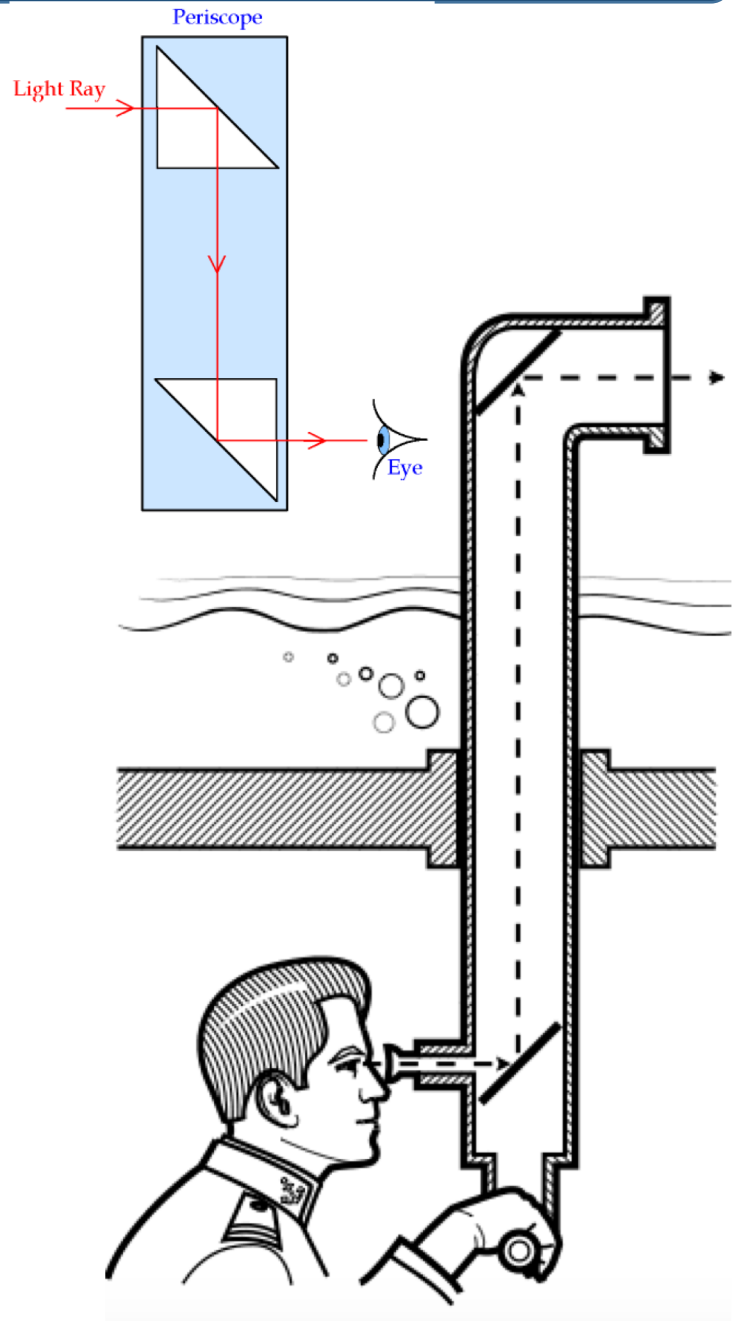




✓ 9/125 μm core/cladding ($n_{clad}$ = 99.9% $n_{core}$)
✓ Charles Kao: Nobel prize in Physics (2009)

# Chapter 1. The nature and propagation of light (2.0h)

## 1.3 Total internal reflection (TIR)

**Exp.1.3: A leaky periscope**
**A submarine periscope uses two totally reflecting 45° - 45° - 90° prisms with total internal reflection on the sides adjacent to the 45° angles. Explain why the periscope will no longer work if it springs a leak and the bottom prism is covered with water.**




- When it springs a leak and the bottom prism is covered with water.

The critical angle for water ($n_b$=1.33) on glass ($n_a$=1.52) is:

$$\sin\theta_{\text{crit}} = \frac{1.33}{1.52} = 0.875$$

$$\theta_{\text{crit}} = 61.0^\circ$$

- The incident angle 45° for a totally reflecting prism is smaller than this new critical angle.

$\Rightarrow$ No TIR at the glass-water interface.

$\Rightarrow$ Most of the light is transmitted into the water, very little is reflected back into the prism.

# Chapter 1. The nature and propagation of light (2.0h)

## 1.4 Dispersion

- **Definition:**

The dependence of wave speed and index of refraction on wavelength.

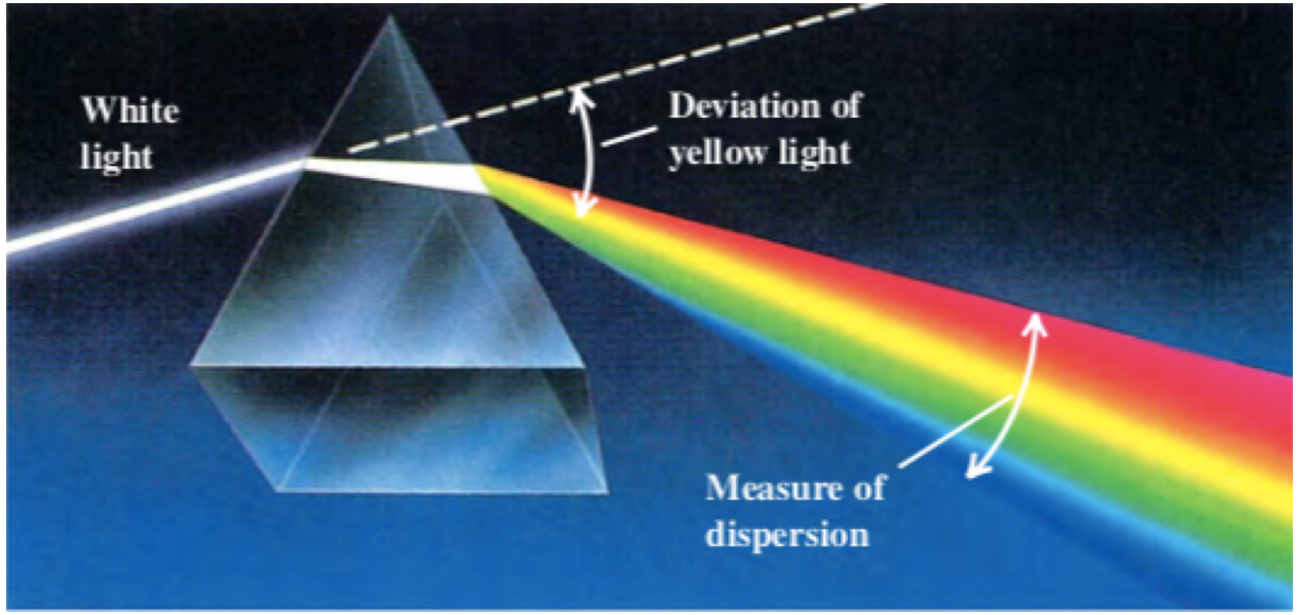
- **Properties:**

✓ For most materials: $n \downarrow (\uparrow)$ with $\uparrow (\downarrow) \lambda$ and $\downarrow (\uparrow) f$.

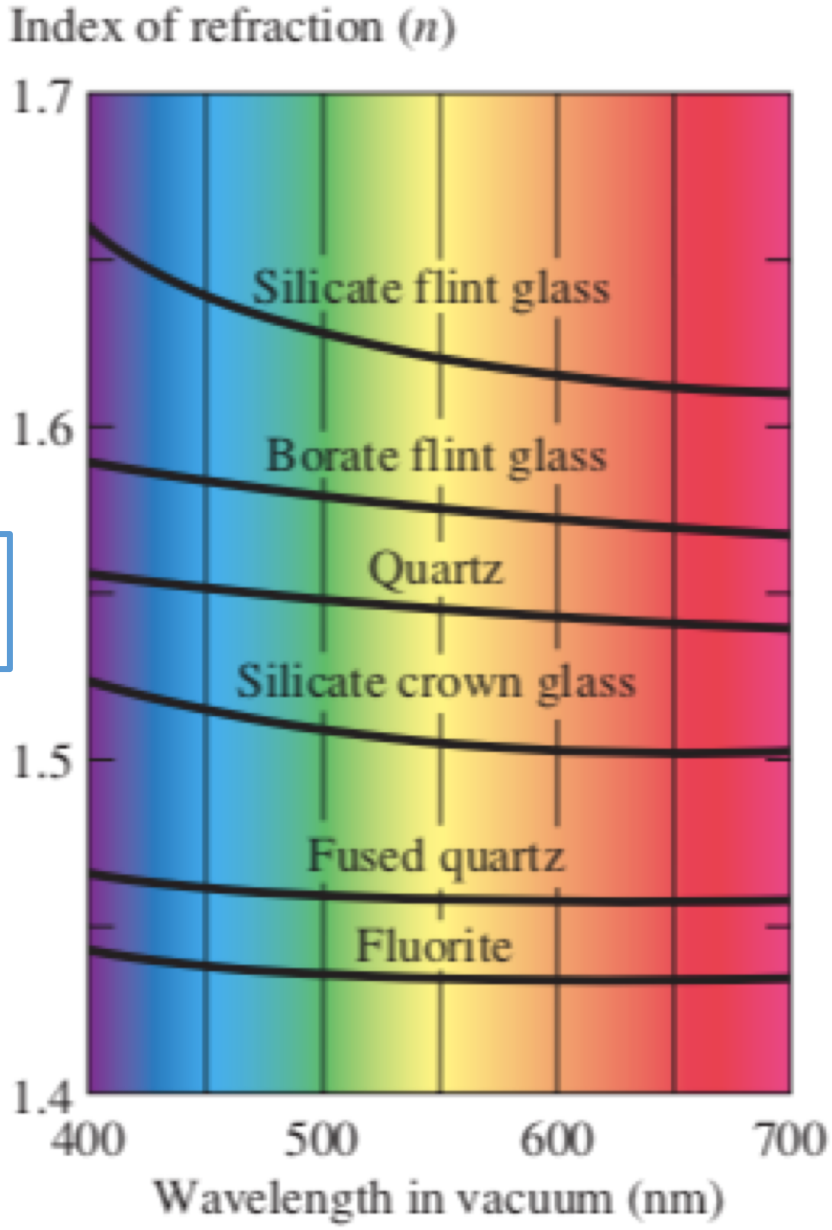
✓ The deviation (change of direction) produced by the prism increases with increasing index of refraction and frequency and decreasing wavelength.

✓ The amount of dispersion depends on the difference between the refractive indexes for violet and red light.

$$\lambda = \frac{\lambda_0}{n}$$



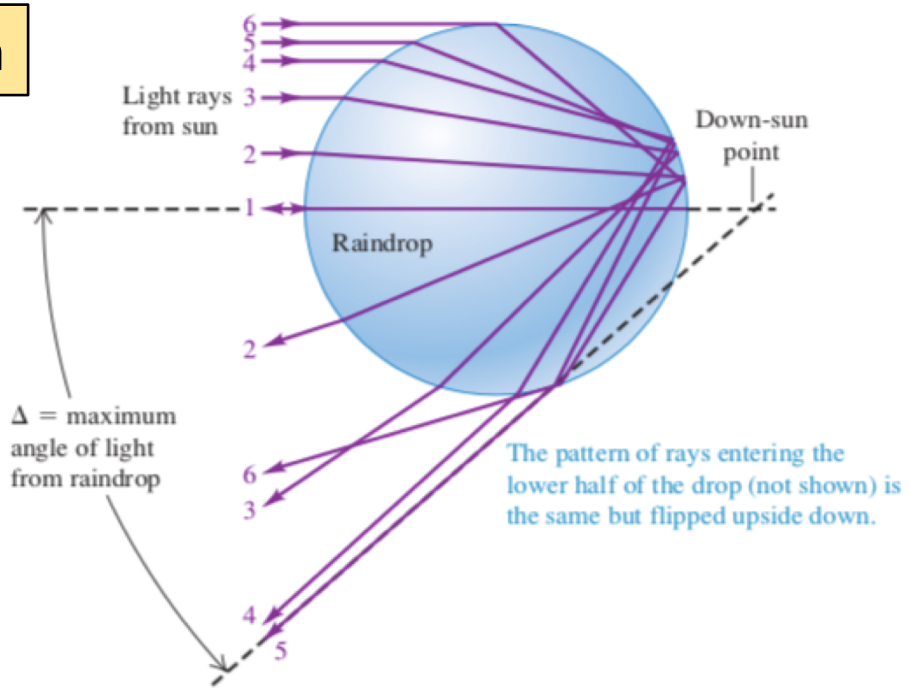

Dispersion of light by a prism. The band of colors: a **spectrum**.

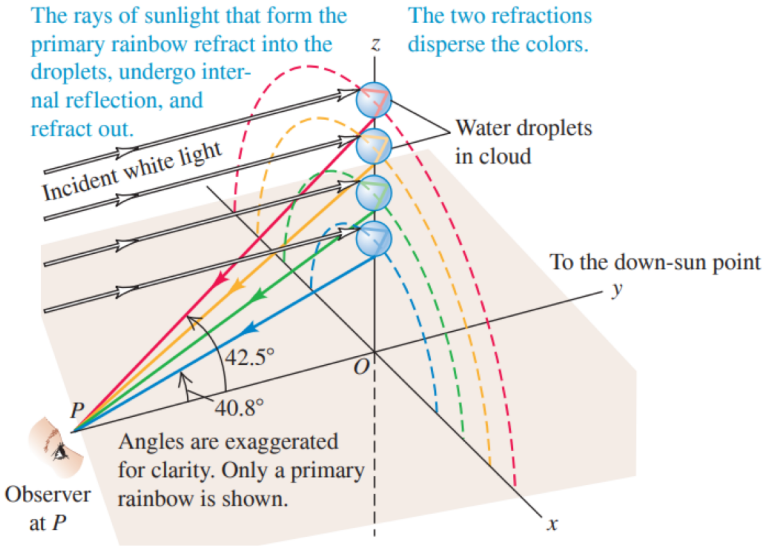


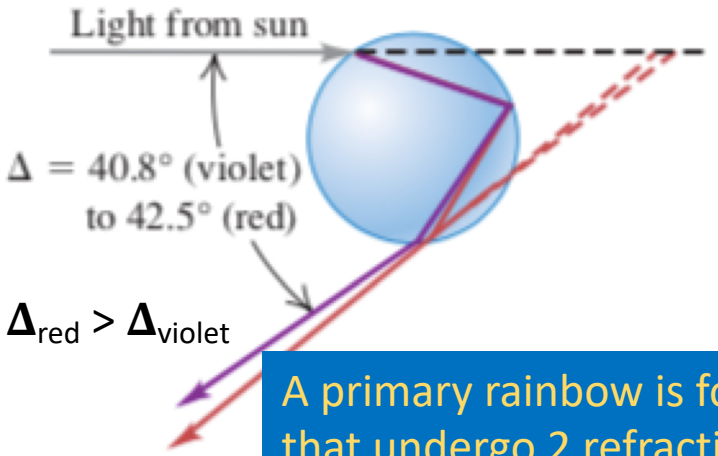
# Chapter 1. The nature and propagation of light (2.0h)

## 1.4 Dispersion

### Rainbows



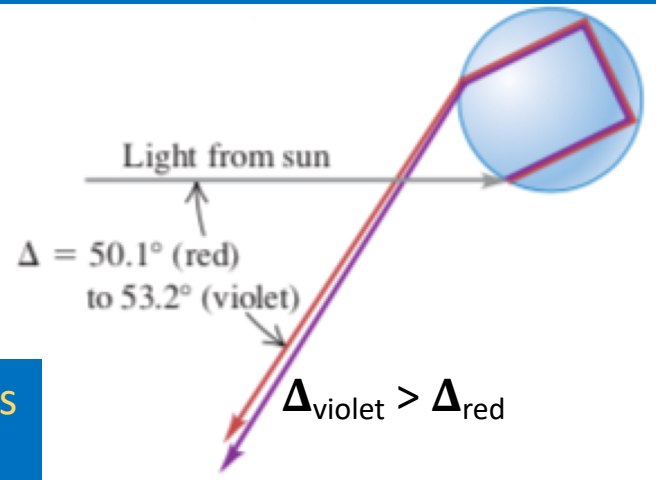



$\Delta_{red} > \Delta_{violet}$

A primary rainbow is formed by rays that undergo 2 refractions + 1 TIR.

A secondary rainbow is formed by rays that undergo 2 refractions + 2TIR.

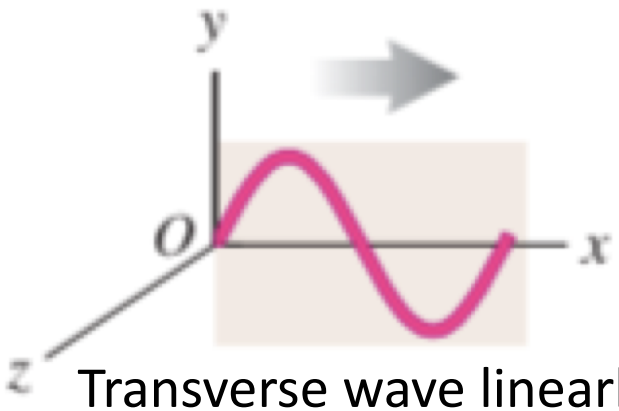



$\Delta_{violet} > \Delta_{red}$

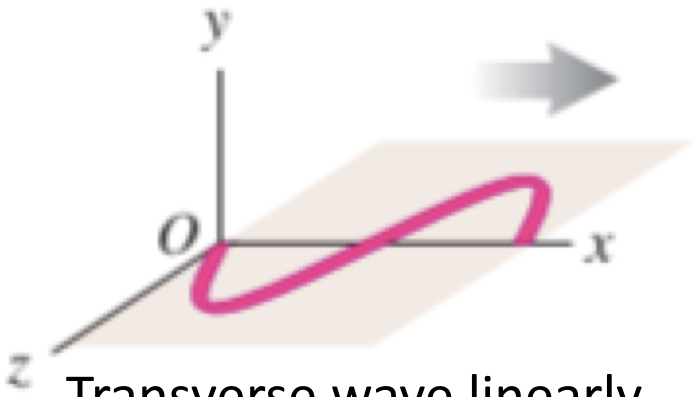
# Chapter 1. The nature and propagation of light (2.0h)

**1.5 Polarization** the direction of ***E*** in an electromagnetic wave

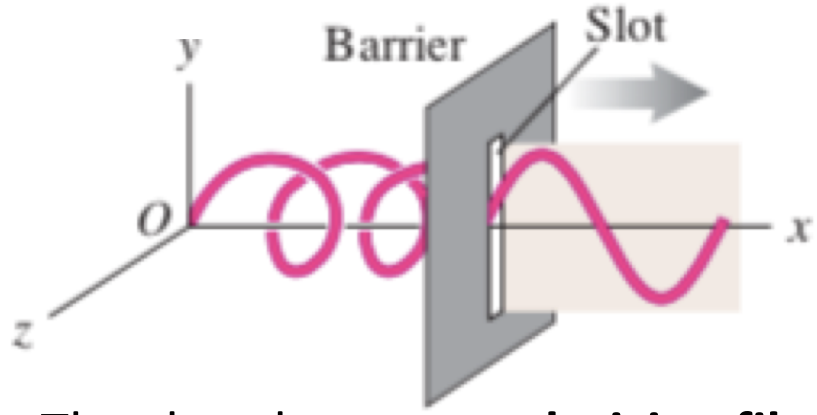
- For transverse waves:




Transverse wave linearly polarized in the *y*-direction.




Transverse wave linearly polarized in the *z*-direction.




The slot plays as a **polarizing filter**, passing only components polarized in the *y*-direction.

- For electromagnetic wave:

$$\boldsymbol{E}(x,t)=\boldsymbol{j}E_{max}\cos(kx-\omega t)$$

polarized in the *y*-direction.

$$\boldsymbol{B}(x,t)=\boldsymbol{k}B_{max}\cos(kx-\omega t)$$

Electrons oscillate vertically, producing vertically polarized electromagnetic waves that propagate away from the antenna in the horizontal direction.

**A radio transmitter**

**Broadcast antenna**

**Visible light!**

**the filament produces unpolarized light waves.**

# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

### 1.5.1 Polarizing filters

- For electromagnetic waves, the construction of polarizing filters depend on the wavelength.

✓ For microwaves ($\lambda$ ~ few cm):
a good polarizer is an array of closed spaced,
// conducting wires insulated from each other.

⇒ *e* are free to move along the length of the conducting wires in response to a wave whose ***E*** // wires

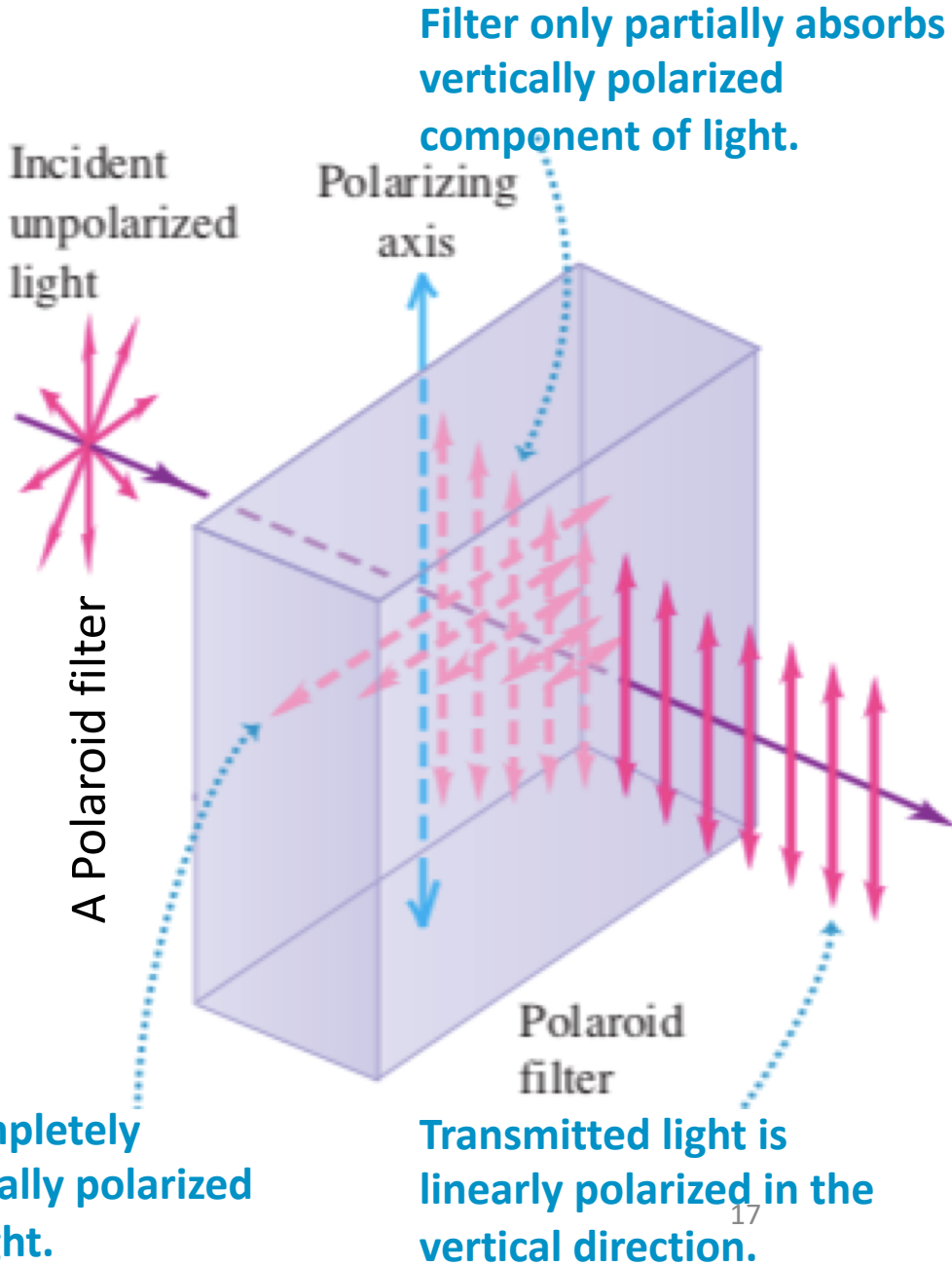
⇒ Waves with ***E*** ⊥ wires: pass through almost unaffected, since *e* can't move through the air between the wires.

✓ For visible light: Polaroid, widely used for sunglasses and polarizing filters for camera lenses.

- A Polaroid filter transmits:

+) ≥ 80% the intensity of a wave polarized // a certain axis in the material: polarizing axis

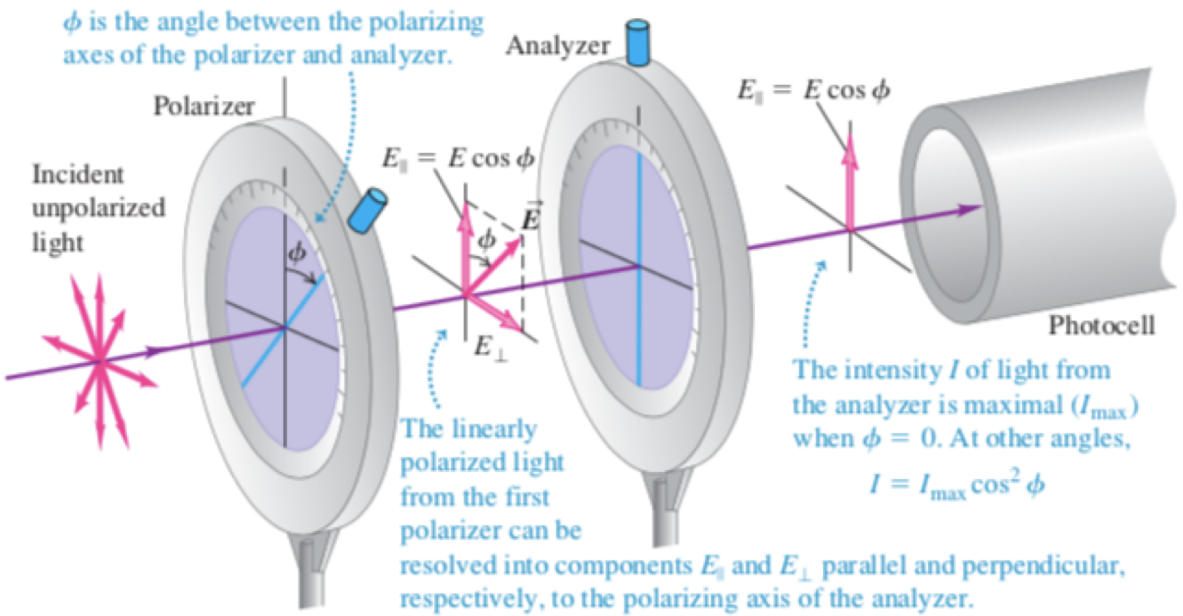
+) ≤ 1% polarized ⊥ to this axis.

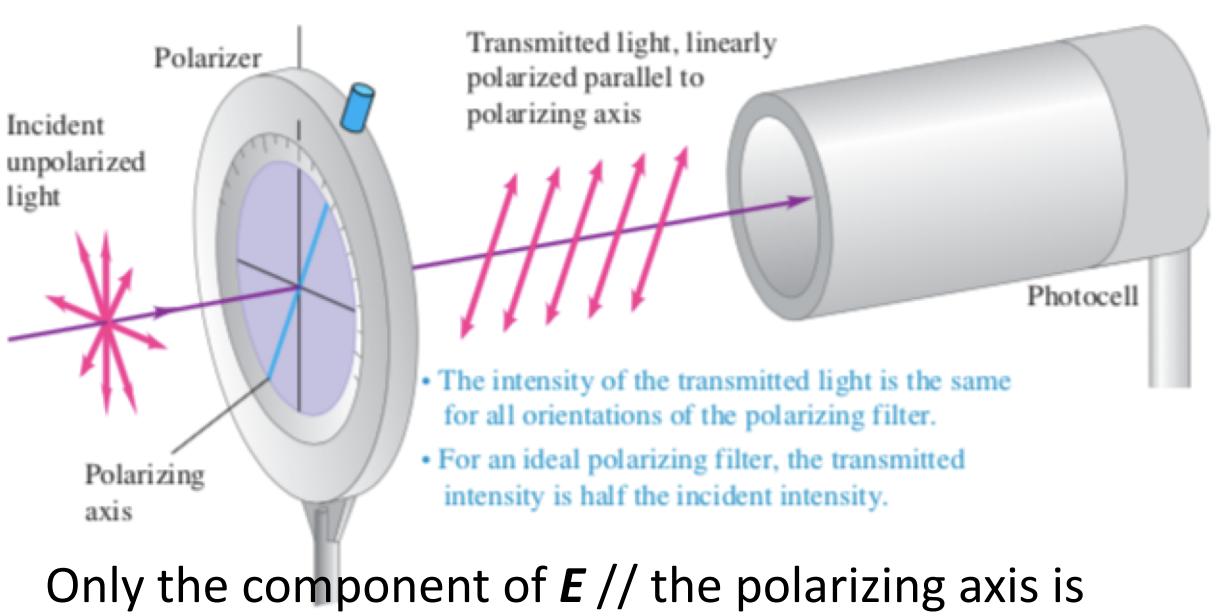


# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

### 1.5.2 Using Polarizing filters




- Only the component of ***E*** // the polarizing axis is transmitted. => The light emerging from the polarizer is linearly polarized // the polarizing axis.
- For an ideal polarizer, the intensity of the transmitted light is *exactly half* that of the incident unpolarized light

- The polarizing axis of the analyzer makes an angle $\phi$ with the polarizing axis of the 1st polarizer.

=> Only the parallel component, with amplitude *E*cos $\phi$, is transmitted by the analyzer.

- The intensity of the polarized light transmitted through the analyzer is

**Malus's** law: $I = I_{\max}\cos^2\phi$

*I*max is the maximum transmitted intensity
*I* is the amount transmitted at angle $\phi$

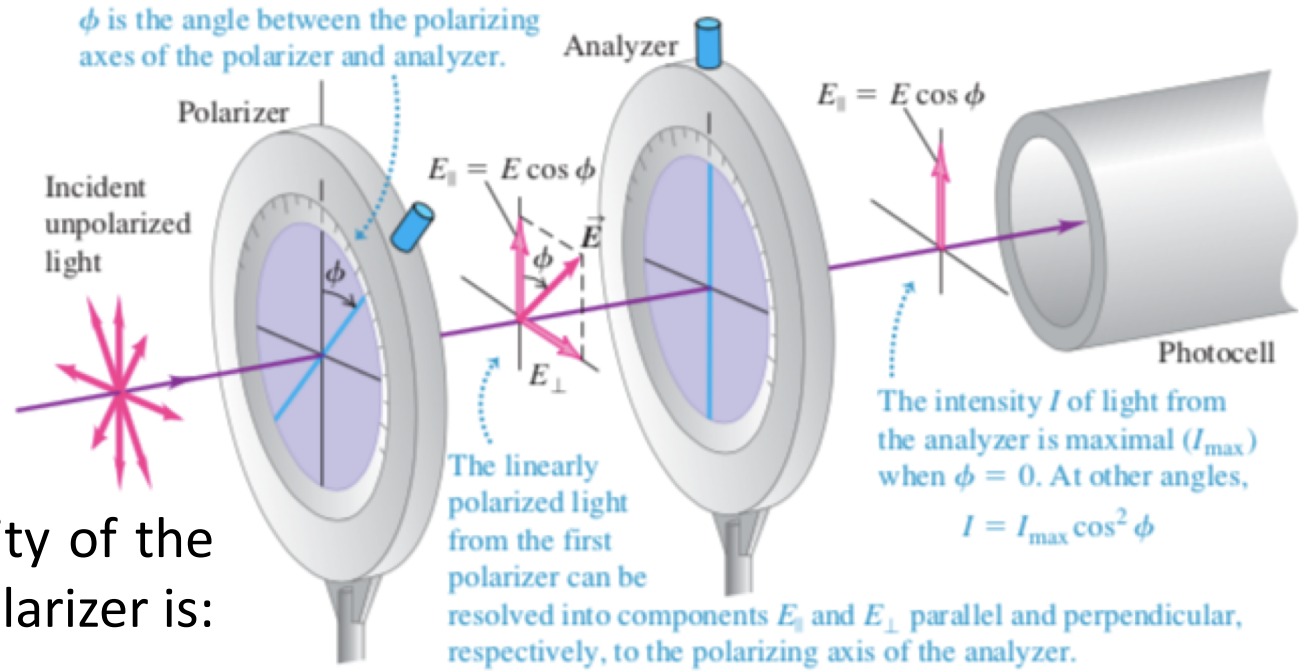
$\phi$=0

$\phi$=90°

# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

**Exp.1.4: Two polarizers in combination**
**In this figure, the incident unpolarized light has intensity $I_0$. Find the intensities transmitted by the first and second polarizers if the angle between axes of the two filters is 30°.**




- The incident light is unpolarized, so the intensity of the linearly polarized light transmitted by the 1st polarizer is: $\frac{I_0}{2}$.
- The intensity transmitted by the second polarizer is

$$I = I_{\max}\cos^2\phi = \frac{I_0}{2}\times\frac{3}{4} = \frac{3}{8}I_0$$

# Chapter 1. The nature and propagation of light (2.0h)

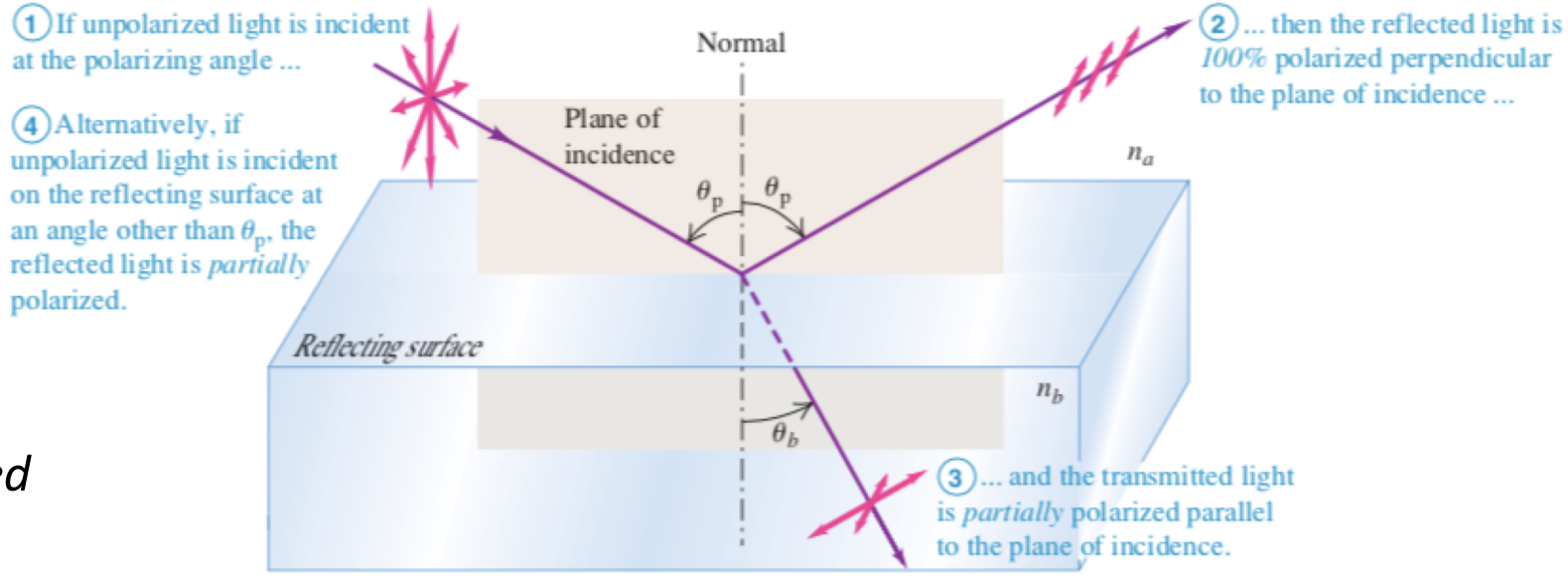
## 1.5 Polarization

### 1.5.3 Polarization by reflection

- **For most angles of incidence**:

waves with ***E*** ⊥ the plane of incidence (// the reflecting surface) are reflected more strongly than those for which ***E*** lies in this plane.

=> The reflected light is *partially polarized* in the direction ⊥ the plane of incidence.




- **At one particular angle of incidence**, the **polarizing angle** $\theta_p$:

the light for which:

+ ***E*** lies in the plane of incidence is *not reflected at all* but is completely refracted.

+ ***E*** ⊥ the plane of incidence is partially reflected and partially refracted.

=> The *reflected* light is *completely* polarized ⊥ to the plane of incidence.

=>The *refracted* (transmitted) light is *partially* polarized // this plane;
the refracted light is a mixture of the component // the plane of incidence, all of which is refracted, and the remainder of the perpendicular component.

# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

### 1.5.3 Polarization by reflection

The angle of incidence: $\theta_{\mathrm{p}}$.

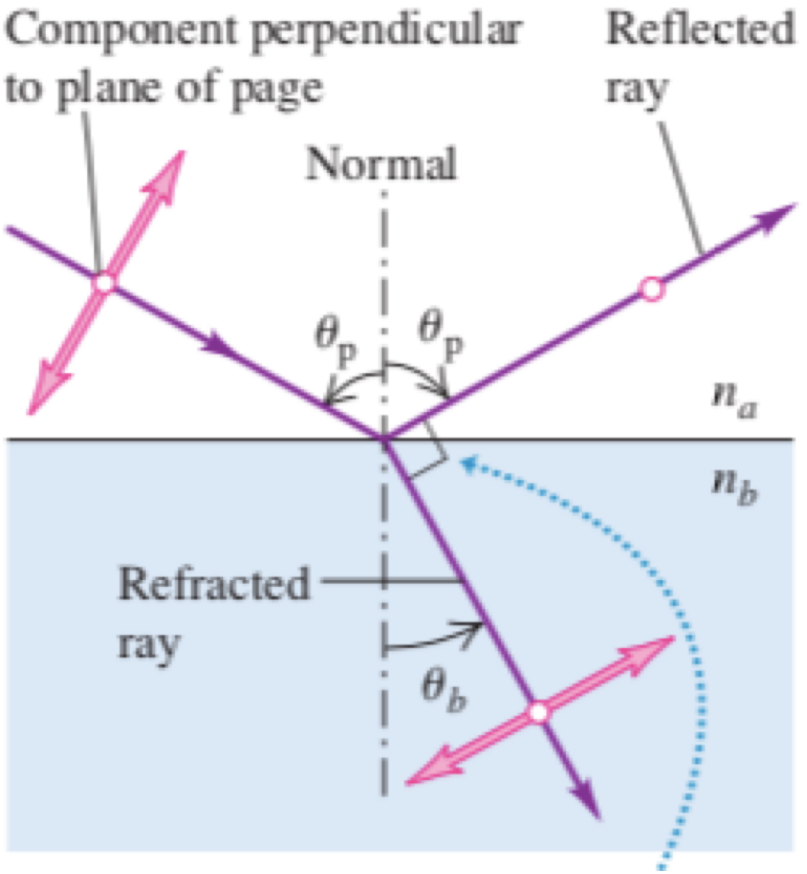
The angle of refraction $\theta_{\mathrm{b}}$: $90° - \theta_{\mathrm{p}}$.

$$\tan\theta_{\mathrm{p}} = \frac{n_b}{n_a}$$ **Brewster's** law for the polarizing angle

$\theta_{\mathrm{p}}$: polarizing angle
(angle of incidence for which reflected light is 100% polarized)
$n_a$, $n_b$: the refractive index of two media *a* and *b*

- **Application:**

  Why are polarizing filters widely used in sunglasses?



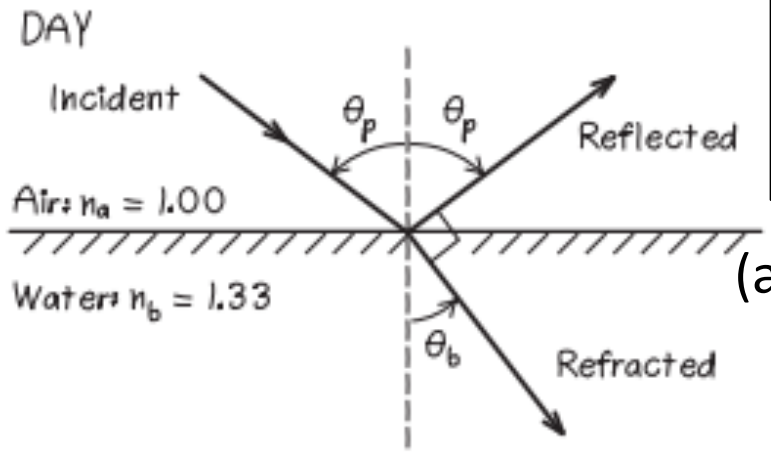

When light strikes a surface at the polarizing angle, the reflected and refracted rays are perpendicular to each other and

$$\tan\theta_{\mathrm{p}} = \frac{n_b}{n_a}$$

# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

### 1.5.3 Polarization by reflection




**Exp.1.5: Reflection from a swimming pool's surface**

**Sunlight reflects off the smooth surface of a swimming pool.**

**(a) For what angle of reflection is the reflected light completely polarized?**

**(b) What is the corresponding angle of refraction?**

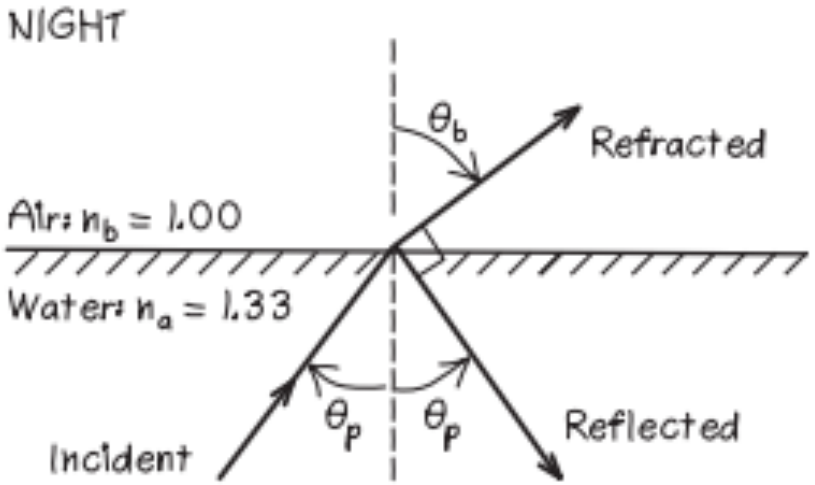
**(c) At night, an underwater floodlight is turned on in the pool. Repeat parts (a) and (b) for rays from the floodlight that strike the surface from below.**

(a) The angle of reflection for that the reflected light completely polarized is

$$\tan\theta_{\mathrm{p}} = \frac{n_b}{n_a} \Rightarrow \theta_{\mathrm{p}} = \arctan\frac{n_b}{n_a} = \arctan\frac{1.33}{1.00} = 53.1^\circ$$

(b) The incident light is at the polarizing angle, so

$$\theta_b = 90^\circ - \theta_{\mathrm{p}} = 36.9^\circ$$




(c) At night,

$$\theta_{\mathrm{p}} = \arctan\frac{n_b}{n_a} = \arctan\frac{1.00}{1.33} = 36.9^\circ$$

$$\theta_b = 90^\circ - \theta_{\mathrm{p}} = 53.1^\circ$$

# Chapter 1. The nature and propagation of light (2.0h)

## 1.5 Polarization

### 1.5.4 Circular and elliptical polarization

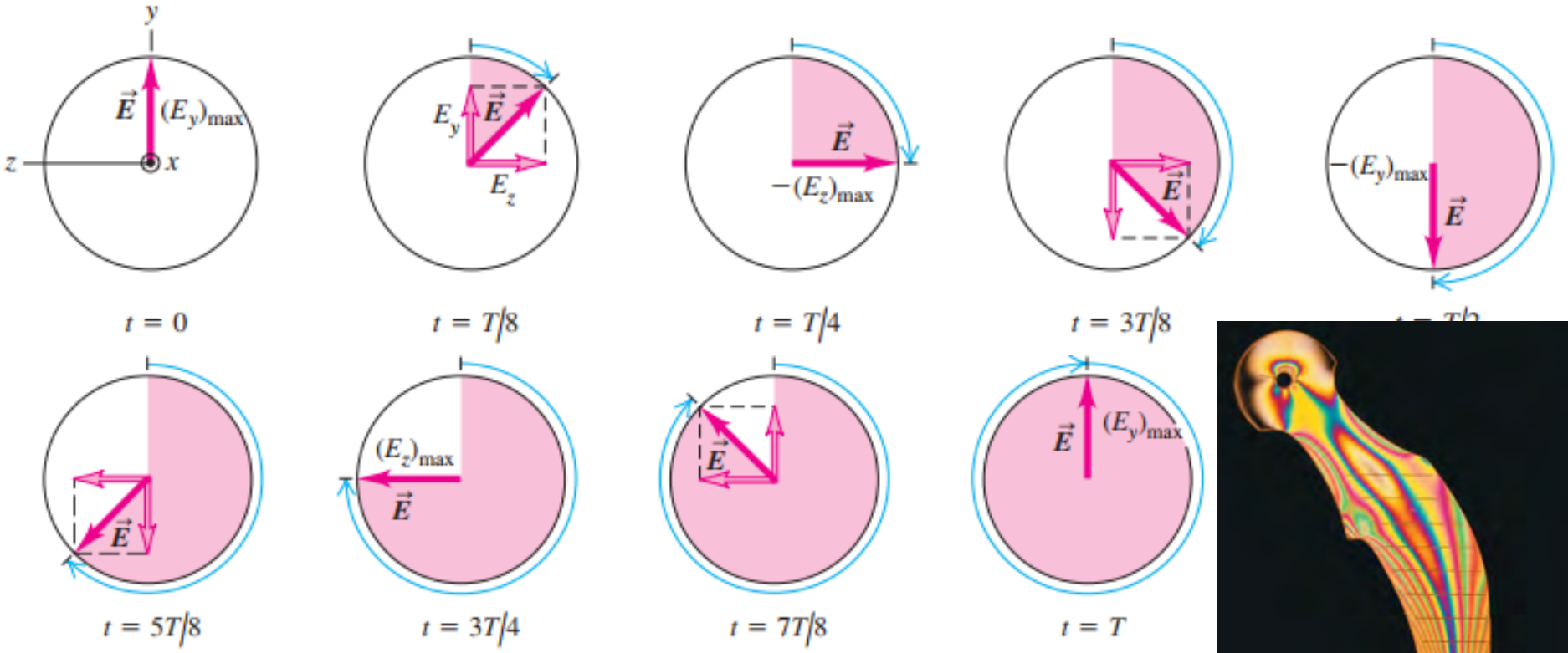
- **Circular polarization** of an electromagnetic wave moving // *x*-axis.

The $E_y$ lags the $E_z$ by a quarter-cycle.
Magnitude of **E** is constant, **E** rotates in a circle.

- **Elliptical polarization:**

✓ If the phase difference between two component waves is not quarter-cycle or,

✓ the two component waves have different amplitudes.

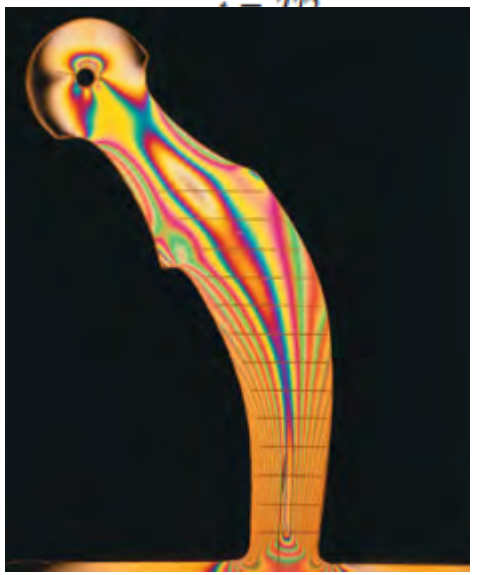



- **Birefringence:**

Different indexes of refraction for different directions of polarization

### 1.5.5 Photoelasticity

Some optical materials that are not normally birefringent become so when they are subjected to mechanical stress.



The plastic model of an artificial hip joint.

# Chapter 1. The nature and propagation of light (2.0h)

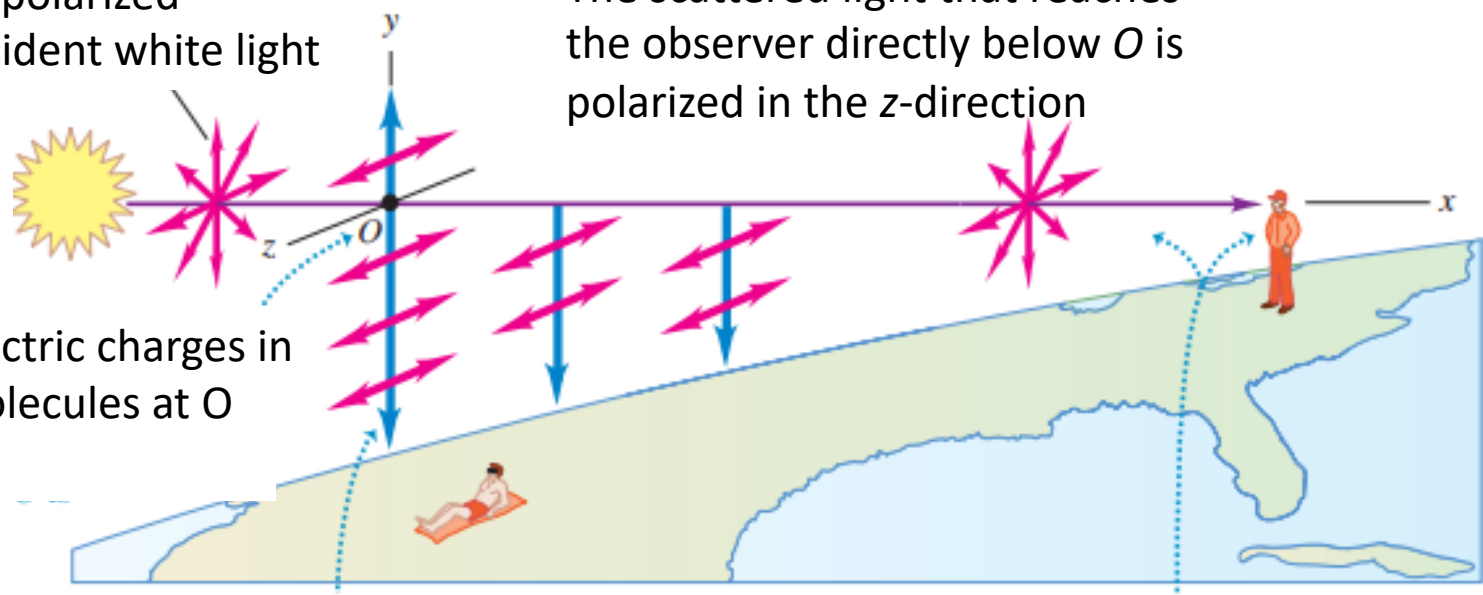
## 1.6 Scattering of light

### 1.6.1 Phenomenon

**Why is the sky blue?**

- Scattering: the sunlight has been absorbed and then re-radiated in a variety of directions.

**Why are sunsets red?**




Air molecules scatter blue light more effectively than red light

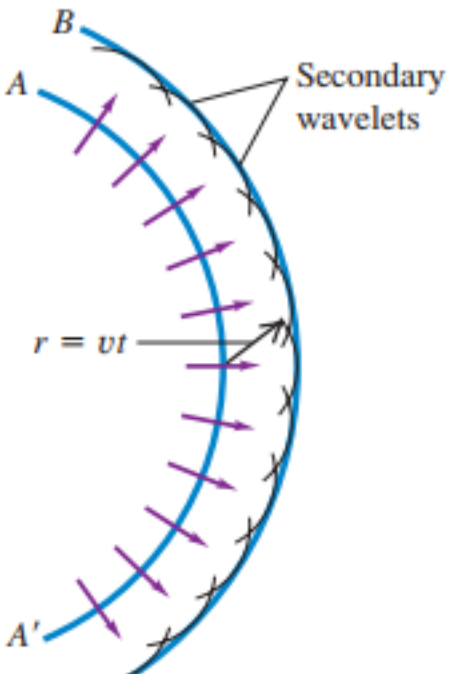
This observer sees reddened sunlight because most of the blue light has been scattered out.

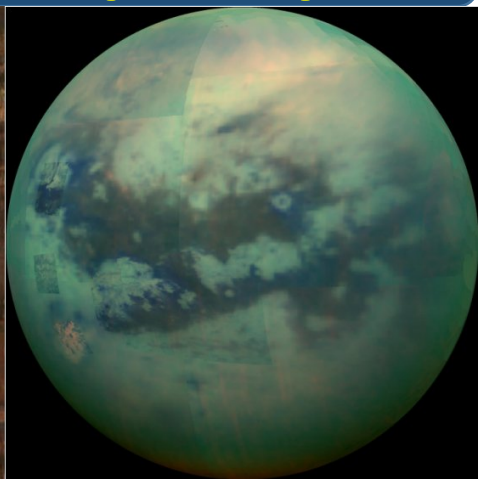
Why do clouds look white?

# Chapter 1. The nature and propagation of light (2.0h)

## 1.7 Huygen's principle

- Every **point of a wave front** may be considered the **source of secondary wavelets** that spread out in **all directions** with a **speed equal to the speed of propagation of the wave**.

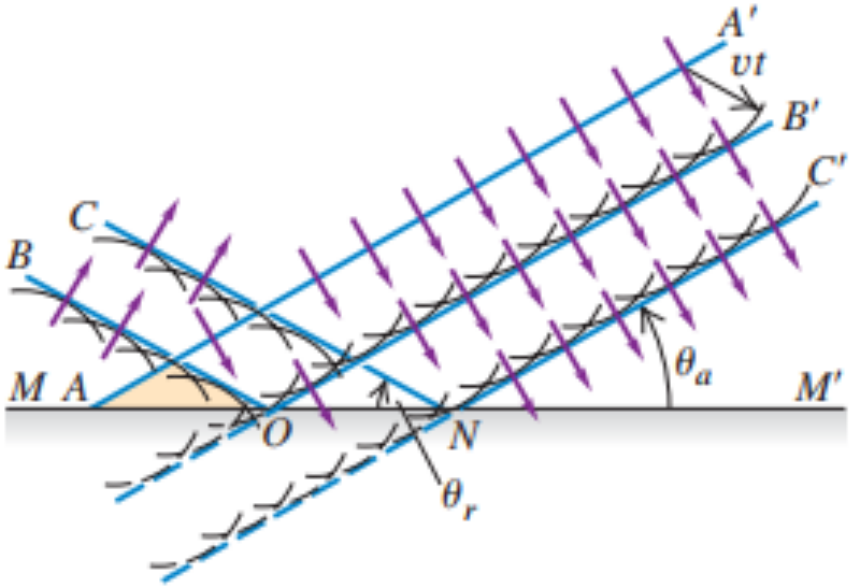


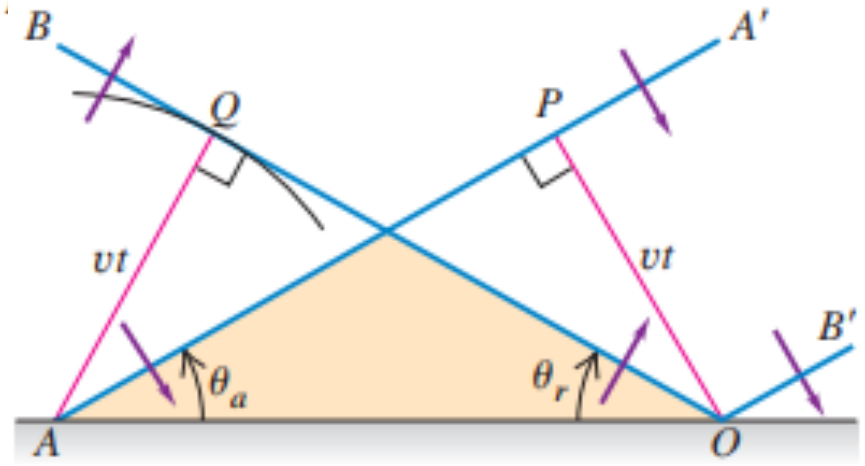


On 25 March 1655 Christiaan Huygens discovered Saturn's satellite Titan.

## 1.7.1 Reflection and Huygen's principle

- Lines AA', OB', NC': successive positions of a wave front approaching the surface MM'



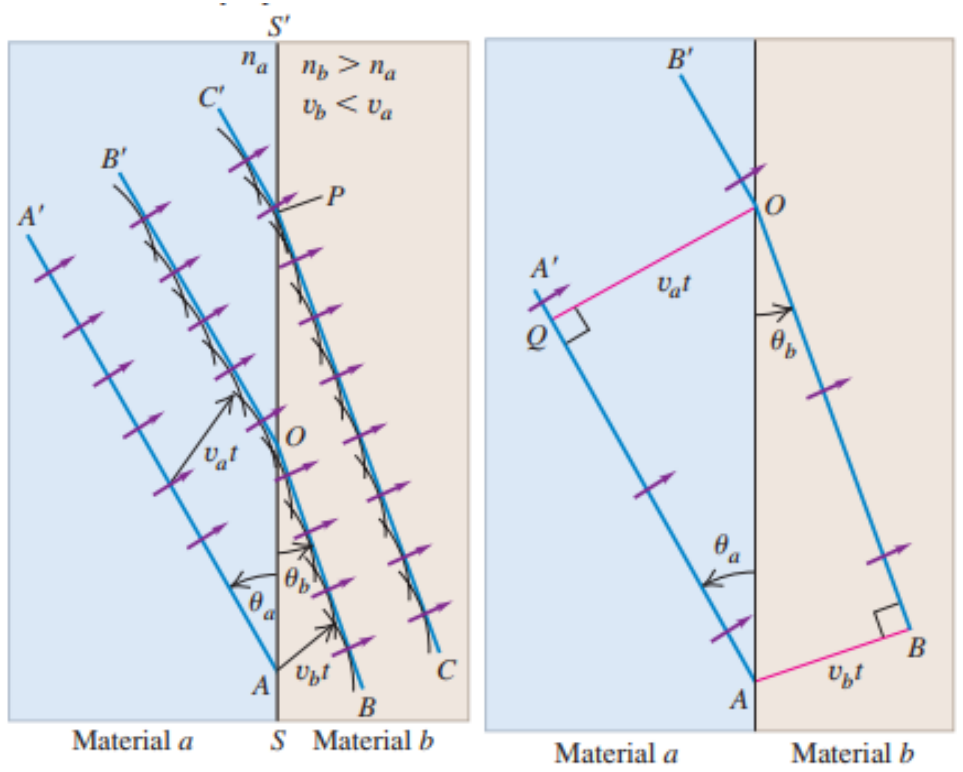

$\theta_r = \theta_a$

# Chapter 1. The nature and propagation of light (2.0h)

## 1.7 Huygen's principle

### 1.7.2 Refraction and Huygen's principle

- Wave front AA' arrives at the boundary surface SS' between two transparent *a* and *b*.



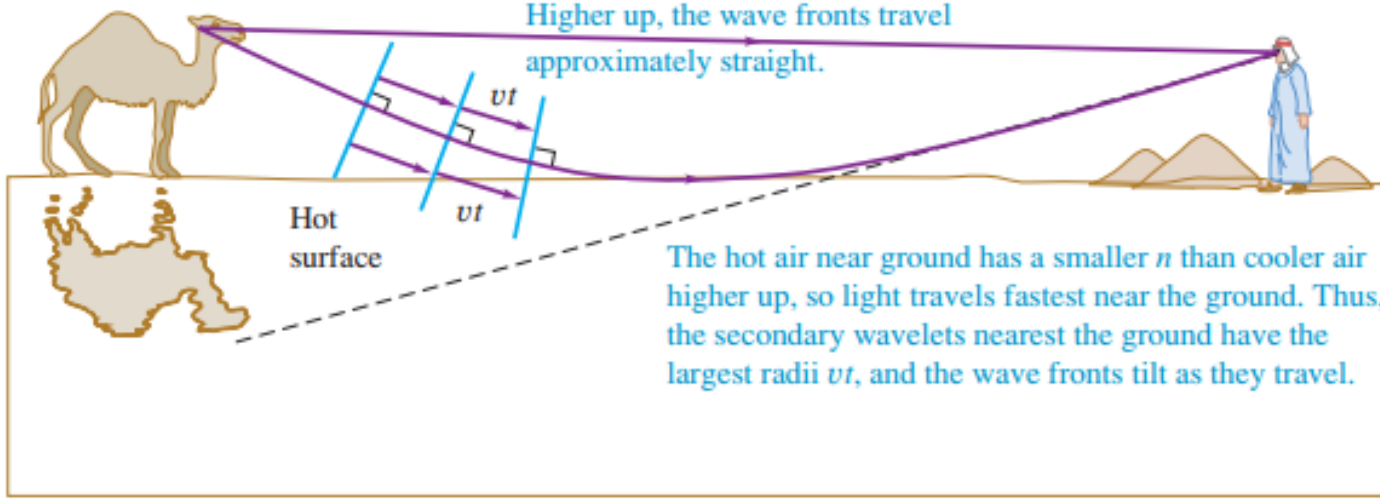

$$\sin\theta_a = \frac{v_a t}{AO} \qquad \sin\theta_b = \frac{v_b t}{AO}$$

$$\frac{\sin\theta_a}{\sin\theta_b} = \frac{v_a}{v_b} \qquad \frac{n_b}{n_a} = \frac{c/v_b}{c/v_a} = \frac{v_a}{v_b}$$

$$\frac{\sin\theta_a}{\sin\theta_b} = \frac{n_b}{n_a}$$

$$\boxed{n_a \sin\theta_a = n_b \sin\theta_b}$$

Mirages???



*“When you really want something the whole universe conspires in helping you to achieve it”*
*(The Alchemist, Paulo Coelho)*

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UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

# GOOD LUCK and THANK YOU

*Make it Simple but Significant!*